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Optimized Feature Weighting Framework for Multi-Document Summarization Using Binary Swarm Intelligence Techniques

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ABSTRACT: Auto-Text summarization is the process through which a compressed version of the original text is created. Extraction-based approaches are the methods through which important sentences and information are extracted from a single or multiple documents. This method selects the sentences based on the calculation of scores for each sentence. The process of extracting sentences that are most important is a trivial task for the text mining researchers. The previous works concentrated on the automatic extraction of features through weighted TF-IDF and dimension reduction using the centroid-based k-means clustering. This paper is intended to apply the extractive methods for automatic text summarization at the multi-document level using swarm intelligence. Particle swarm optimization is used in this research for identifying the optimal weight of the feature scores for differentiating between the relevant and irrelevant features. The ROUGE tool is used for the evaluation. The student data provided by the educational department of the US is considered for the experiment. The summary that are generated using the proposed methodology is compared to those of other methods, such as Genetic Algorithm and Ant Colony Optimization, and the TF-IDF: CBC as the benchmark method. The results from the experiment prove that the extracted summaries produced after the integration of the proposed method into TF-IDF: CBC perform better.

KEYWORDS: Text Summarization, k-means clustering, TF-IDF, Particle Agent Swarm Optimization, ROUGGE, and GA-ACO.

1 Introduction

In the digital era, most of the information needed from a user is available as a large source of unstructured data. This unstructured data is normally found as large pieces of text. The volume of data being accumulated is increasing rapidly with respect to time. The need for automatic summarization of these large textual data has become vital for the efficient extraction of information [1]. Summaries that are man-made are termed as manual summarization, and these involve the process of reading and understanding the text and thereafter highlighting the important information. On the other hand, the summaries that are generated by machines are termed automatic summarization. The essentiality of automatic summaries is very much in demand owing to the size and complexity of these large texts. The overheads, such as information overload, time complexity, and accuracy, that are faced by manual methods of summarization have made the auto-summarization methods more popular and vital. Text summarization involves the process of bringing in the important information extracted from a large piece or pieces of text. It takes a longer time and more human power to understand these texts and to extract the most relevant information [2]. Hence, auto-summarization methods are largely preferred in domains such as email categorization, news summary, article summarization, and research works etc.

The earlier effort on an automatic text summarization system, that further developed in the late 1950s, involved selecting important sentences from the original document and concatenating them into a shorter form. Automatic text summarization techniques are classified into different approaches. Some of these techniques are classified based on the input document used for the summary. Single-document summarization uses only one document to produce a single summary, while multi-document summarization uses many documents that are related to the same topic to create a single summarization. The summarization methods are broadly classified into extractive and abstractive summarization [3]. The former method consists of selecting the most important sentences from the source document and concatenating them into a shorter version. The latter is the technique where at least a few of the entire material is purposefully left out in the input [4].

The task of evaluating the quality of a summary is very important. The evaluation can be assessed manually and/or automatically. Manual evaluation is done by humans or automatically by special tools. Two categories of methods used in text summarization are extrinsic and intrinsic [5-6]. Extrinsic evaluation measures the efficiency and acceptability of summaries in some tasks based on the idea of how useful the summaries are for a given task, for example, reading comprehension or relevance assessment. On the other hand, intrinsic evaluation measures a summary quality of the system itself by comparison to some gold standard, such as human-generated summaries. The evaluation can be assessed manually and/or automatically. Manual evaluation is done by a human. The Recall-Oriented Understanding for Gisting Evaluation (ROUGE), for example, is an automatic intrinsic evaluator of summary systems for Document Understanding [7].

One of the easier ways for the determination of optimum feature weight is through exhaustive research, in which the subset of features is evaluated with the universal set of all features. This approach is not practical even for a feature set of a medium size. The feature extraction normally involves heuristic search methods to avoid the complexities. However, the optimality of the end subset is normally reduced [8]. Some of the methods that have attracted more researchers are the Genetic Algorithms, ACOs, and PSOs that collect better solutions using the knowledge obtained from previous steps. The genetic method of optimization is based on the natural selection process. They are operated similarly to natural genetics for supervising the searches in the given space [9].

Owing to its advantages, GAs is normally used as a tool for feature extraction in text mining. PSO was introduced by Kennedy [10] and is based on the social and psychological behavior of swarms in nature. The particles in the swarm follow a simple behavior that kindles the success of the neighboring particles. The outcome behavior of the entire group identifies the optimum regions in a high-dimensional space.

This research work proposes a Particle Agent Swarm Optimization-based text summarization technique for identifying the best fit weight in TF-IDF: CBC for multi-document auto text summarization. The evaluations are done through the ROUGE toolkit for performance measures. The results from experiments proved that the summaries generated after the integration of the proposed method into TF-IDF: CBC are better than other algorithms. The rest of the paper is organized as follows. Section 2 gives the gist of the related works in the PSO-based summarization. Section 3 elucidates the problem statement, and Section 4 deals with the proposed methodology in detail. Section 5 describes the experimental setup and the evaluation measures. Section 6 depicts the results obtained from experiments and discusses the same. Section 7 gives the conclusion and opens research problems that can be taken for future experiments.

2 Related Works

The PSO method is used in the current research as a machine learning technique for feature extraction and to study its effect in the optimum weight calculation of the features. The scores are combined with the features that are produced by the PSO in the proposed method for auto-text summarization. Only a few studies have been carried out with the PSO method on the text summarization problem, and the important studies are as follows:

PSO was successfully applied to some related problems, like text classification. The particle swarm optimization was also applied successfully in the feature selection problem. PSO was implemented for selecting the subset of features for a classification and to train the neural network [11]. The classification problem of the extracted features is addressed using the PSO method [12]. PSO was adopted for the feature extraction for the enhancement of performance of the SVMs and the neural network for obtaining an optimum solution to the e-learning texts [13]. PSO was deployed with a Support Vector Machine for the determination of parameters and to improve the accuracy of classification [14].

According to the successes of PSO in the above studies, the PSO method was deployed to investigate the impact of feature extraction on its structure of the summarized text [15], where the features used are divided into two types: “complex” features and “simple” features. Complex features are “sentence centrality”, “title feature”, and “word sentence score”; simple features are “keyword” and “first sentence similarity”. After computing each feature score, the PSO was used to identify which features are more effective. The score of ROUGE-1 was used to calculate the value of the fitness function. The dataset used for training the system comprised one hundred articles from DUC-2002. The PSO parameters were initialized, and the “gbests” values were computed to extract the weight of each feature. Results showed that complex features received higher weights than did simple features, which indicates that feature structure plays an important role in the feature selection process. To assess the results, the authors installed one human model summary as a reference and a second as a benchmark. The MS Word summarizing tool and the initial human-made summary are compared. The results proved that the PSO method performed better than the Microsoft Word tool and achieved accuracy closest to a human-tailored summary. The maximal margin relevance [16] (MMR) method is proposed to enhance summary diversity.

3 Problem Definition

Features are the most important entities in the summarization of texts. Considering that all the features are equal will result in the generation of a poor summary. Building an optimal feature weighting mechanism for high-quality summary generation is considered a complex task. PSO was proposed before to solve this problem, but when the same is surveyed, especially text summarization based on evolutionary and swarm intelligence algorithms, it is found that previous works established an unfair comparison criterion to test which evolutionary and swarm algorithm is better. In this study, the criteria are unified to judge which algorithm is better in reality for implementing a text summarizer application.

4 Proposed Methodology

The operational framework of this research consists of four phases: Phase 1: Feature extraction through TF-IDF: CBC, Phase 2: Binary Particle Swarm Based Text Summarization, Phase 3: Training Procedure, and Phase 4: Testing Procedure. The architecture is shown in Figure 1.

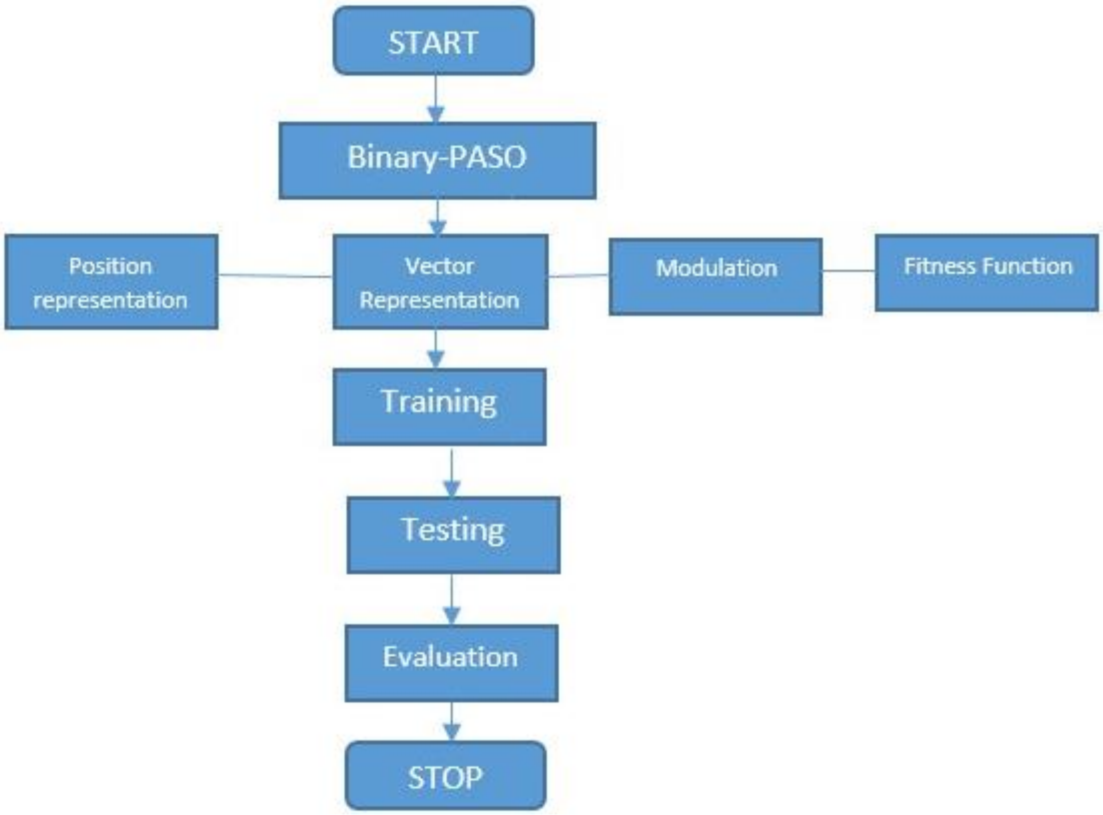

Figure 1: Proposed Architecture

4.1 Binary Particle Swarm-Based Text Summarization

The feature scoring is considered the base of the text summarization process. The particle swarm optimization (PSO) is used as a machine learning method to learn the feature weight from the training data. The extracted weights are used to adjust the feature scores.

4.1.1 Particle Position Representation and Configuration

The Binary-PSO is considered, where the position of the particle is represented as a binary string. Every bit takes the value of either 0 or 1 and represents individual features. The bit value of 1 indicates that a feature is selected and unselected otherwise. The initial bit represents the first feature, and the second bit represents the second feature, and the same is followed throughout. The pictorial representation of the same is provided in Figure 2.

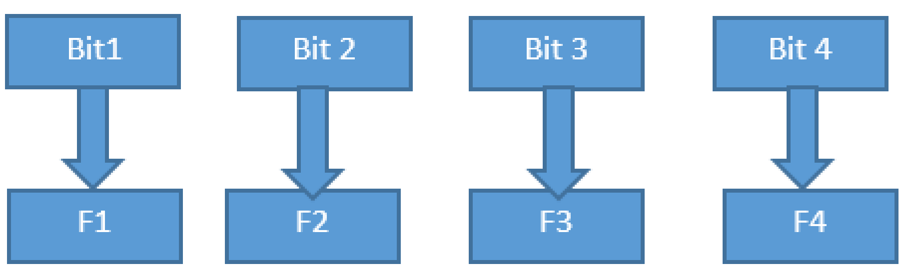

Figure 2: Bit representation of features

4.1.2 Particle Velocity Representation and Configuration

The velocity of the particle is depicted in the same manner as the position of the particle, wherein every value of the bit is retrieved using the sigmoid function. The velocity used in the sigmoid is represented in equations (1) and (2),

$$Vi(t+1) = w.Vi(t)+c1.r1(t).[Pi(t)-Xi(t)] + c2.r2(t).[Pg(t)-Xi(t)] \tag{1}$$

$$Xi(t+1) = Xi(t)+Vi(t+1) \tag{2}$$

In which c1 and c2 represent the positive constants and are called learning rates, and $Pi = (pi1, pi2, \ldots, piD)$ indicates the swarm's previous best position. r1 and r2 are the values taken randomly from the range [0,1]. Xi = (x_{i1}, x_{i2}, . . . , x_{iD}) shows the position belonging to particle (i) in the space with dimension D. t denotes the number of iterations, and the weight of inertia is denoted by w. The best particle from the entire population is denoted by the index g, and $Vi = (vi1, vi2, \ldots, viD)$shows the positions change with respect to time. If the summation pertaining to the factors present in the RHS of equation (1) exceeds a certain threshold, the velocity of the particle on every dimension is then clamped at the maximum velocity V_{Max}.

4.1.3 Binary Modulation formula

Particle Swarm Optimization algorithm is one of the Swarm Intelligence algorithms. The Swarm Intelligence algorithms are used with real values. To enable PSO to search in a binary space, we need a modulation formula in order to modulate real values into binary values.

$$x_{jk(t+1)} = \left\{ {}^{0}_{1} if\ P_{jk}\ (t) \geq \frac{1}{1+\exp\left(-v_{k)}(t)\right.} \right\} \tag{3}$$

In this research, we used modulation as in equation (3) to perform this task or 0 otherwise.

4.1.4 The Fitness Function

The fitness function is used as a measuring unit in optimization techniques. The fitness function is used to determine which particle obtains the best solution and is considered the fittest value. This value will change if the new particle generates better fitness than the previous particle. Equation (4) shows the fitness function determination. The function to represent the fitness is obtained through the least degree of the saw particle, which is jumping off from the path, and the finest change in the strength of the particle in the degree of the equation.

$$Fitness(X_i) = \phi.\gamma(S_i(t)) + \varphi.(n - |S_i(t)|) \tag{4}$$

In the above equation, $Si(t)$denotes the subset of the feature identified by the particle I in an iteration t,with the length given as($Si(t)$), the fitness is calculated to measure the performance of the classifier ($Si(t)$)with the length of the subset. Φ and φ are the couple of parameters which control the relative weight of the classifier and the length of the subset features denoted by ϕ _[0,1] $and\ \varphi = 1 - \phi$. This indicates that the performance of the classifier and the feature subset have contrasting effects on the features that are extracted. In this research, more importance is given to the classifiers' performance rather than the length of the subsets. Hence, the same is set as $\phi = 0.8, \varphi = 0.2$.

4.2 Training procedure

In this study, the PSO used 70 documents from the US education department. In the beginning, all the documents are dealt with initial pre-processing of sentence segmentation, tokenization, and removal of stopwords. Then, the Weighted TF-IDF method is used to extract the features [18]. The scores of each sentence are then represented using the vectors as in Table 1, and the results of the scores obtained as used as the input for the scoring function of PSO as in equation (5).

Table 1: Sample feature scores

Sentence / Feature	TF	SL	SP	ND	TW	Total
S1	0.68	0.57	0.42	0.24	0.81	2.72
S2	0.48	0.47	0.28	0.67	0.74	2.64
S3	0.27	0.42	0.68	0.81	0.42	2.60
S…	…	…	…	…	…	…
S10	0.38	0.42	0.48	0.61	0.74	2.63

$$Score\ (xi) = \sum_{j=1}^{5} s(fj) * Vopp(i) \tag{5}$$

The (s_i) denoted the score of the sentence and s_i, $S(f_{j)}$ represents the score for the feature F_j and $Vopp_i$ represents the value of the $bit(i)$ in the present position after calculating the scores of all the documents using the equation (5), and are ranked in a descending manner, and finally, the $top\ n$ sentences are selected to frame the summary, wherein n denotes the length of the summary, which is pre-defined. The proposed method uses the threshold value of 20% of the size of the source text as the summary length. The proposed PSO makes use of the equation as in (4.5) for selecting n sentences from the source for composing the summary. The summary that is generated is again used as the input for the fitness function. The evaluation of the summary is made, and the p_{best} is identified, which represents the value of the evaluation of the best summary that is generated by that particular particle. The g_{best} is also calculated, which indicates the value of the evaluation of the best summary that is generated by the particle in the search space.

By the end of an iteration, the particle's position that has the value of g_{best} will be selected as the vector for the best selected feature in every document. The final feature weights are calculated over the vectors of the feature weights of all documents in the data collection.

$$Length\ of\ Summary = \frac{Ratio\ defined\ by\ user * Length\ of\ the\ document}{100} \tag{6}$$

4.3 Testing procedure

The goal of employing the PSO is to find and optimize the corresponding weight w_j of each feature f_j. Equation 7 calculates the feature weights,

$$Score - Weight\ (si) = \sum_{j=1}^{5} wj * Vopp(f_{j\ (si)}) \tag{7}$$

Where score weights (s_i) is the score of sentence s, w_j is the weight of the feature j produced by PSO, j is the number of features, and is a function that calculates the score of the feature j. The testing procedure used 30 documents from the DUC2002 data set. The testing procedure is begin with input document, then implementing the preprocessing process (segmentation, tokenization, remove stop word and stem the word), then extracting features for each sentence, then modify the score of each feature based on the features weights that produced in training process, the score of each sentence is then calculated for all documents as in equation (7) followed by the ordering of these sentences in an descending manner and to select the $top\ n$ sentence and re-ordering to the original order of the source document is done. The ROUGE evaluation measures are used for performance evaluation.

5 Experimental environment and metrics

This research is based on the functional approximation (random research) approach using an intelligent swarm algorithm named "particle swarm optimization". The PSO algorithm is provided with a learning approach (feature weighting). The PSO used to obtain appropriate feature weights. We have set the particle swarm optimization parameters as follows: number of particles=30, maximum number of iterations=500. The commonly used information retrieval metrics of precision, recall, and F-Score. The summary that is generated by humans is the best choice for evaluation. Therefore, the generated summaries in this study were evaluated and compared with the human-generated summaries.

Precision (P) is calculated as the number of sentences that are intersected in-between the summary generated by the system and that of the human, divided by the number of sentences in the summary generated through the system. It is calculated using equation (8).

$$Precision = \frac{|System\ Based\ Summary| \cap |Human\ Based\ Summary|}{|System\ Based\ Summary|} \tag{8}$$

Recall denoted by R is the number of sentences that are intersected in-between the summary generated by the system and the summary generated by a human, divided by the number of sentences in the model. It is calculated using equation (9).

$$Recall = \frac{|System\ based\ Ummary| \cap |Human\ based\ Summary|}{|Human\ Based\ Summary|} \tag{9}$$

The F-score maintains the performance of the system based on the “Precision” and “Recall” values and is calculated using equation (10).

$$F = \frac{2 * Precision * Recall}{Precision + Recall} \tag{10}$$

5.1 Rouge: method of evaluation

The ROUGE (Recall-Oriented Understudy for Gisting Evaluation) is an evaluation system for the measurement of the quality of the summaries in comparison with the human-created summaries [17]. The ROUGE tool depends on counting n-gram co-occurrences in the system summary and in the reference summary. ROUGE provides four different measures, namely ROUGE-N, ROUGE-L, ROUGE-W, ROUGE-S, and ROUGE-SU.

5.2 Rouge-N

ROUGE-N measures co-occurrences of n-grams. The ROUGE-N score can be calculated as:

$$Rougue - N = \frac{\sum (Gramn)_n \in S * \ Count_{Match}(gram)_n}{\sum (Gramn)_n \in S * Count(gram)_n} \tag{11}$$

Where S is the sentence taken as a reference, n denotes the length of the gram, and $Count(match)$ is the $n - grams$ that are shared between a given set of reference summaries, and the $count(gram)$ is the total count of the $n - grams$ in-between the set of summaries that are present in the system-generated summary.

5.3 Results and Discussions

This chapter describes the results of the applied PSO algorithm in text summarization and compares the generated summary after applying feature weights with other algorithms, which are the Ant-colony algorithm and Genetic algorithm; the Differential Evolution algorithm is used as a benchmark. The H2-H1 Compression method is used as a benchmark too; this method is produced by comparing the human summary called (H2) with the reference human summary called (H1). This compression is established to evaluate the summary against human performance. These algorithms used the same five statistical features (Title Feature, Sentence Length, Sentence Position, Numerical Data, and Thematic Words) and the same data set. The ROUGE packet is used to evaluate the obtained results. When implementing the testing process, we used the ROUGE-N evaluation measure. ROUGE-N measure counts all occurring (shared) words. The generated summaries by these algorithms (PSO, GA, ACO) are compared with the DE algorithm summary. Tables 2 and 3 compare the three methods using ROUGE-1, ROUGE-2. Average recall (avg-R), average precision (avg-P), and average F-measure (avg-F) are calculated for each method. Figures 1 and 2 visualize the same results obtained.

Table 2: Results based on ROGUE 1

Method	Performance Measures		
	Avg-P	Avg-R	Avg-F-Score
H2-H1	96.1	96.8	96.8
DE	87.4	86.8	86.8
ACO	81.2	84.6	82.5
GA	91.2	92.4	93.4
PASO	95.1	96.2	94.8

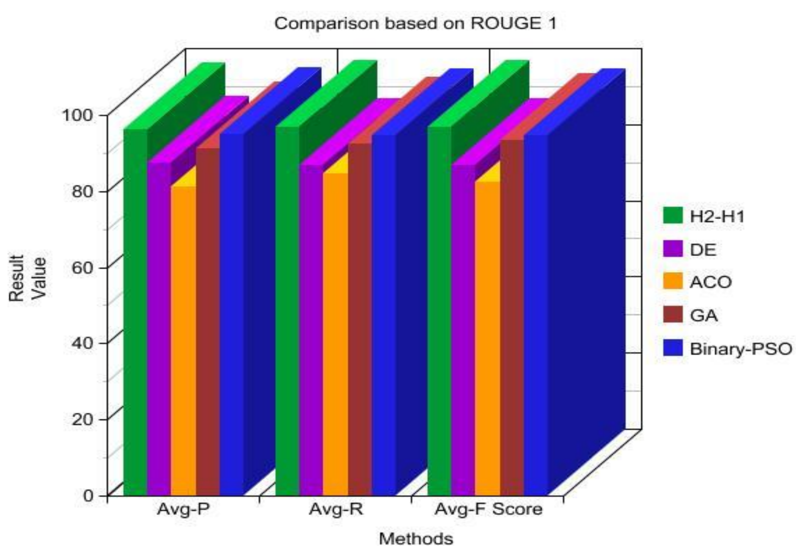

Figure 3: Comparative results – ROUGE 1

Table 3 shows the results obtained from the ROUGE 2 evaluation. It is seen that the proposed method has the closest proximity to handwritten summaries when compared to all the other methods under discussion. Figure 4 depicts the same in a graphical format.

Table 3: Results based on ROGUE 2

Method	Performance Measures		
	Avg-P	Avg-R	Avg-F-Score
H2-H1	97.2	96.8	97.4

DE	89.2	86.6	84.8
ACO	84.7	83.8	87.4
GA	90.4	91.7	90.9
PASO	96.4	95.2	96.4

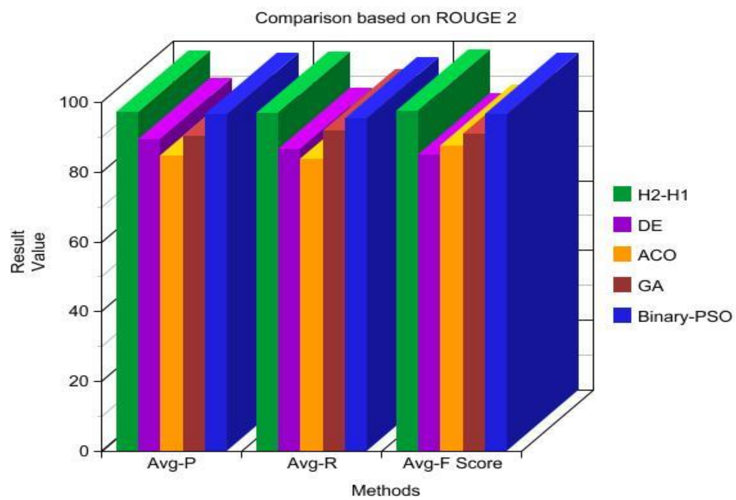

Figure 4: Comparative results – ROUGE 2

6 Conclusion

The performance of feature-weighting in automatic text summarization has been proven to generate high-quality summarization. In this research, binary PSO was used to obtain feature weights. The standard version of PSO used real values for the position and velocity of the particles. The modulation function is used to modulate the real values into binary values to determine the inclusion of features for weighting. Binary PSO is used to determine a better particle for each document. The structure of the particle includes five bits; each bit represents one feature. After the end of PSO iterations, the average weight of each feature is calculated, called the feature-weight. The proposed method is applied to the TF-IDF model, and the generated summary in this stage is compared with other algorithms (GA, DE, and ACO). The experiments were conducted on the large dataset from US educational records, and the evaluation was done using the ROUGE tools. The experiments prove that the proposed methodology performs better than other Genetic Algorithms and ACOs.

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